

# Toward a Computational Natural Law Defense of the Sanctity of Human Life

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Respect for human dignity, often phrased as the sacredness, or transcendent worth, or sanctity of human life, lies at the bedrock of Catholic Social Teaching (hereafter, CST)<sup>1</sup>. Whether as a thinner account based on rational agency<sup>2</sup>, a fuller account based on additional requirements of prosperous communities<sup>3</sup>, both or somewhere in between<sup>4</sup>, a natural law defense of human dignity has to make some appeal to rational criteria available in some sense to all human beings; that such criteria are available for this appeal also lies at the bedrock of CST<sup>5</sup>. The phenomena of seemingly intractable moral disputes about fundamentals of the natural law calls into question the availability of such universal rational standards. Alasdair MacIntyre defended an account of such standards in support of CST, aided by both critical and supportive commentators, in 2009 (*Intractable Disputes About the Natural Law*, IDNL, ed. Cunningham), which was the University of Notre Dame's response to then Cardinal Ratzinger's earlier plea to address this very challenge to CST<sup>6</sup>. In what follows that defense will be presented as discussed in MacIntyre's reply to his interlocutors. It will be shown to fall short as a defense so far as MacIntyre has presented it, but to hold promise in the direction toward which it points. The argument will then build much beyond what Prof. MacIntyre advocated in 2009, in a way that

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<sup>1</sup> A nice summary of the roots of human dignity in Scripture and Tradition from the US Conference of Catholic Bishops is found here:

<https://www.usccb.org/beliefs-and-teachings/what-we-believe/catholic-social-teaching/life-and-dignity-of-the-human-person> (accessed 11/8/2021).

<sup>2</sup> See Alasdair MacIntyre, "Intractable Moral Disagreements" and "From Answers to Questions: A Response to the Responses" in *Intractable Disputes About the Natural Law* (IDNL), e. Cunningham, Univ. of Notre Dame Press, 2009.

<sup>3</sup> See Jean Porter, "Does the Natural Law Provide a Universally Valid Morality?", IDNL p. 53 - 95.

<sup>4</sup> See Thomas Loughran, "Resanctifying Human Life" in *Recovering the Sacred: Catholic Faith, Worship and Practice* (Proceedings of the Twelfth Convention of the Fellowship of Catholic Scholars) ed. Williams, Northeast Books, 1990

<sup>5</sup> IDNL p. 341.

<sup>6</sup> This approach is reaffirmed in MacIntyre's subsequent arguments in *Ethics and the Conflicts of Modernity* (ECM), Cambridge University Press, 2016.

comes nearer to what he proposed in his 10<sup>th</sup> Gifford Lecture in 1988.<sup>7</sup> In particular, I will argue that an adequate defense of the sanctity of human life in the natural law tradition as upheld in Catholic Social Teaching requires an integration of the natural sciences in a new master science, which for reasons to be explained later I'll call computational natural law. I defend this claim in three stages: by I) showing that the best current defense (in a tradition led by MacIntyre) is inadequate by its own standards; II) identifying an additional level of justification—that of effective, tradition-specific rhetoric—required and sufficient to produce an evidentiary signal at a yet-unspecified rate; and III) sketching a design of an accelerator/detector system adequate for producing and observing such signal from the interaction of traditions at a practicable rate. That interaction will become the experimental component of a new computational master science of natural law.

## *I.*

To begin, MacIntyre's defense of the rationally obligatory status of the precepts of natural law as articulated by Aquinas will be reviewed, and an important way that the defense falls short will be identified.

MacIntyre grounds his defense in an account of the nature of practical reason as articulated in the Aristotelian-Thomistic tradition. Practical reason, the pursuit of the good, commits us to social deliberation about how to order the most general areas of

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<sup>7</sup> *Three Rival Versions of Moral Inquiry (3RV)* UNDP 1990, Lecture X. See also *ECM*, p. xii.

the good: wealth, pleasure, health, power, honor, and the like. Such deliberation, MacIntyre argues, requires as a prerequisite respect for a series of norms concerning those with whom we do or might deliberate. As to content, these norms include respect for the life, property, honest information, etc. of our interlocutors; as to form, they must apply universally to all, be the same in content for each, operate prior to any inquiry, and be non-inferentially grasped.<sup>8</sup> These norms amount both to what Aquinas articulates as the precepts of the natural law,<sup>9</sup> and to the inviolable respect for human beings captured by the notions of human dignity and respect for the sanctity of human life, conforming with and in fact serving as the foundations of Catholic Social Teaching. At a later stage in MacIntyre's argument, a parallel case is constructed for extending these norms to the goods of sexuality and family life, completing a one-to-one correspondence with Aquinas's account of the precepts of natural law.<sup>10</sup>

MacIntyre's defense depends upon acceptance of Aquinas's account of practical rationality, and there are other and rival accounts. Disagreements about specific requirements of natural law will turn on acceptance of its precepts, and that acceptance on how we judge between rival accounts of the nature of practical rationality. Such disagreements concern the highest principles of inquiry, admitting of no appeal to further shared principles, and are thus incommensurable. Yet if contending participants

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<sup>8</sup> *IDNL* p. 24-25; See McKenny's summary of MacIntyre's argument, "Moral Disagreement and the Limits of Reason" also in his *IDNL*, p. 210.

<sup>9</sup> MacIntyre is aware that he goes beyond the texts of Aquinas, but asserts that he knows of "no cogent account of Aquinas on these matters that is not compelled to such an interpretation. *Whose Justice? Which Rationality (WJ?)* University of Notre Dame Press, 1988, p. 180.

<sup>10</sup> *IDNL*, pl. 338-9.

in such a dialogue can accomplish what MacIntyre acknowledges to be the “difficult exercise” of entering into imaginative participation in the traditions of their rivals as second-first-language speakers--and such a possibility can never be ruled out<sup>11</sup>--they can come to see one another’s positions from each other’s perspectives. These remain different perspectives with different commitments. Yet it is possible that from both perspectives, some same tradition of the two--both now fully visible from both perspectives--shows itself to have resources to resolve its problems and explain the lack of any similar ability of its rival, to the satisfaction of the mature members of that rival tradition.<sup>12</sup> Thus agreement might be achieved for complementary sets of reasons from incommensurable perspectives, and relativism thus overcome, in such an encounter between rival traditions. In this way, MacIntyre points the way toward securing a rational foundation for precepts of the natural law, in spite of even intractable moral disagreement. For it remains possible that such differently-motivated agreement lies down this difficult road, even while adherents of rival traditions find themselves unable or unwilling<sup>13</sup> to make the necessary arduous climb into second-first-language competency in the Aristotelian-Thomistic tradition necessary for the journey. Gerald McKenny, whose account of MacIntyre’s argument MacIntyre himself describes as “remarkably lucid and accurate”, calls the necessary effort, Herculean.<sup>14</sup> But that journey is nonetheless possible, and possible with it, the rational accessibility of the primary

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<sup>11</sup> *3RV* p. 120: “...the possibility of learning to understand the other incommensurable point of view from within imaginatively, before it can be occupied intellectually, can never be ruled out.”

<sup>12</sup> *IDNL*, p. 34-5; see also McKenny’s explication in *IDNL* of MacIntyre’s argument on this same point, p. 207.

<sup>13</sup> *IDNL* p. 37; cf *WJ?* p. 396.

<sup>14</sup> *IDNL* p. 207.

precepts of the natural law as defended within the so-far-still-standing

Aristotelian-Thomistic tradition. So far, MacIntyre's defense.

It can further be said of MacIntyre's defense, although he himself is appropriately modest enough not to say it, that such Herculean effort has been shown to be not just possible but actual in MacIntyre's own body of work. With that full corpus as unspoken background, MacIntyre offers just a single example of such an effort in his recounting of the history of Utilitarianism. As a 2nd-first-language speaker of the Utilitarian tradition, MacIntyre tells their story as one of a history of commitments that it fails to keep: of providing coherent accounts of both 1) Utilitarianism's greatest good for the greatest number as a rational standard; and 2) some appropriate metric for when such a standard is met, whether in terms of happiness, or fulfilled desire or informed desire, or any other candidate. Moreover, he shows from Thomistic-Aristotelian resources why these problems arose and were destined to remain unsolved, and argues that no similar unresolved difficulties arise for Aquinas's account. He thus attempts to establish in actual fact the existence of an asymmetry that persuades himself and presumably other adherents of the Thomistic Aristotelian tradition who are also 2nd-first-language speakers of Utilitarianism. This is no small feat, showing both that the Herculean ascent into such a competent comparison is not just possible but can actually be made, and that an asymmetry of resources can be seen between the two traditions from mature speakers of both.

Or at least, it is shown that such an ascent can be made into Utilitarianism, in particular, by mature members of the Thomistic tradition, and that such an asymmetry of resources is apparent to those who make it. But in that additional precision lies a challenge for MacIntyre's defense. For what MacIntyre does not show is whether a complementary ascent can be made by a Utilitarian, and that if it were, that a mature Utilitarian reading of history would be sufficiently aligned with his own as to produce agreement that Utilitarianism has failed in a way which Thomistic Aristotelianism has not, and thus that previously intractable disputes between Utilitarianism and Thomistic Aristotelianism are no longer in force and no longer threaten the rationally-accessible natural law foundations of Catholic Social Teaching. Imagine, as Gerald McKenny asks in his rejoinder to MacIntyre, that a tradition "...not only refuses to accept the claim to rational vindication put forth (by its rival) ... but also puts forth its own claim to rational vindication against that tradition?"<sup>15</sup> Doesn't such refusal to agree re-pose the relativist challenge, which on MacIntyre's own account cannot be settled by appeal to neutral standards? It seems that the threat of relativism remains.

The premise MacIntyre needs in order to unseat that threat, beyond what he has already established, is that the judgment of a mature, Thomistic, second-first-language speaker of Utilitarianism would sufficiently align on the crucial points of history with the

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<sup>15</sup> *IDNL*, p. 210.

mature, Utilitarian, second-first-language speaker of Thomistic Aristotelianism. But MacIntyre does not seem to believe that premise, and he offers no defense of it.<sup>16</sup>

One could supply some resources for such a defense from what he has already argued: after all, to be a second-first-language speaker of a tradition is to produce a grasp of it that passes inspection by the mature members of that tradition. So mature imaginary participation in one tradition by a mature member of another is perhaps sufficient to craft an initial argument for the dialectically-established rational supremacy of one tradition over another. But if we grant that much, we can nonetheless still deny that such one-sided imaginative participation is sufficient to judge the success of the argument crafted, from the perspective of the Utilitarians: you may imaginatively lead them to the water in this way, but you can't make them actually drink.<sup>17</sup>

Especially good reason to deny the sufficiency of one-sided historical accounts of any tradition's supremacy regarding the overcoming of relativism comes from behavioral

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<sup>16</sup> Here MacIntyre must defend himself against what amounts to his own charge: that he is like the wrong sort of Thomist, who "never on any major issue refuted their opponents in such a way as to persuade those opponents." *WJ?*, p. 176.

<sup>17</sup> It is assent of the fully-informed Utilitarian from her own perspective to the historical argument crafted imaginatively from a Utilitarian perspective by the 2nd-first-language speaker of Utilitarianism from the Aristotelian/Thomistic tradition—the act of "drinking" in this sense—that constitutes the evidentiary signal of the kind necessary for overcoming relativism between these two traditions, on MacIntyre's own account of dialectical engagement. Yet nothing in MacIntyre's argument in *INDL* or *ECM* provides any explicit identification of that signal. Some historico-empirical evidence of arguably-mature utilitarian converts to a Thomistic perspective is precisely what is required and precisely what is lacking. It is perhaps up to MacIntyre or historians of his intellectual journey to argue whether MacIntyre's own conversion to the Thomistic-Aristotelian tradition constitutes such a conversion; it would, insofar as he represented a mature Utilitarian at some point in his journey and converted from that perspective to Thomism.

neuroscience.<sup>18</sup> A number of popular accounts exist of the measured tendency of each of us to preserve our primary predictive models. Notable among these are Tavis and Aronson's *Mistakes Were Made, But Not By Me*, and the many others cited in the published remedy for these now widely recognized forms of ever-pressing cognitive misbehavior in Galef's *The Scout Mindset: Why Some People See Things Clearly, and Others Don't*. The bottom line of these accounts is that our brains energetically would to preserve our primary interpretive model, a model that functions to predict into our perceptual space a set of awarenesses that must be abruptly interrupted by any sensory data that don't fit. Such interruption triggers a desperate attempt to explain away the outliers. In that sense, we broadcast our own perceived world, something of a

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<sup>18</sup> Contemporary neuroscience reveals that our brains are compulsive model builders and model projectors: the world as it is imminently expected to appear to us is something like a stable hallucination that our brain is spring-loaded to produce. That other traditions are less appealing; that dissenting individuals fail to understand our own perspective well enough to count as disagreeing epistemic peers; that their dissent fails to count as evidence against our account: all these claims that support our prevailing predictive model, our brains are hard-wired to accept. It is literally painful—energetically expensive—to admit contrary evidence, let alone change our minds; the world to which we look for evidence is not neutral ground but rather literally a projection of our own model. An overview of this understanding of the brain is popularly available now in Jeff Hawkins's *A Thousand Brains: A New Theory of Intelligence*; the implications of this emerging general neuroscientific consensus across a great many sectors of human behavior are chronicled in Carol Tavis' and Elliot Aronson's *Mistakes Were Made, But Not By Me*; an account of the chemical balancing involved in Anna Lembke's *Dopamine Nation: Finding Balance in the Age of Indulgence*; and a craft of rational habit formation in light of these neuroscientific deliverances conveyed in Julia Galef's *The Scout Mindset: Why Some People See Things Clearly, and Others Don't*. That these tendencies in human behavior are likely to be explained by large compilations of genetic features at the  $10^3 - 10^4$  SNP level—as all behavioral tendencies seem to be—is the implication of Robert Plomin's *Blueprint: how DNA makes us who we are*. These lines of research provide grounds for healthy suspicion of one's own judgments about the relative strength of our arguments, and of our historical analysis of how their strengths came to be and have endured. We should instead expect that other views of the world have more to be said for them than our present assessment suggests even after dialogue, and employ a variety of strategies to strengthen that expectation, as Galef's work suggests. Translating these strategies to the present argument suggests giving equality of place to the judgment of the 2nd-1st-language speaker of our own tradition. Only when the judgements of persons formed to maturity in both of two traditions, but as inhabitants of different traditions of the two, both agree from their different models that the same tradition of the two has the stronger case, is evidence provided that relativism is overcome in the encounter between these rival traditions. Conversion is an essential element of the evidentiary signal.

stable, controlled hallucination<sup>19</sup>, that is contradicted only with difficulty. Against this background, it is no surprise that Thomists find their own set of histories to favor a Thomistic outcome; their brains are on fire with predictive activity making it so. This conclusion, motivated by these popular works and supporting literature, serves to more deeply reinforce a point MacIntyre has himself made clearly enough<sup>20</sup>: that formation to maturity in a rich and socially-embodied tradition such as Thomism is not the same as imaginative participation in it<sup>21</sup>, and confrontation between traditions requires participation by mature members from both perspectives who have each made the ascent into maturity in the rival tradition in question. So no Thomist is by him or herself qualified to assess the all the relevant histories from all the relevant perspectives--Thomistic history of Utilitarianism and Thomistic history of Thomism, and Utilitarian histories both of Utilitarianism and of Thomism--so as to judge which has the upper hand in their dialectical engagement. Yet it is precisely agreement from both perspectives, and not just about Utilitarianism's capacity to solve its own problems but its relative promise for doing so compared to Thomism's analogous capacity, that

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<sup>19</sup> See this review of work by Anil Seth and Andy Clark:

<https://www.technologyreview.com/2021/08/25/1032121/brains-controlled-hallucination/> ; see also [https://www.edge.org/conversation/andy\\_clark-perception-as-controlled-hallucination](https://www.edge.org/conversation/andy_clark-perception-as-controlled-hallucination), both accessed 11/9/2021

<sup>20</sup> See especially *WJ?* p. 395.

<sup>21</sup> See *3RV*, p. 117: "It is not that the adherent of one particular standpoint cannot on occasion understand some rival point of view both intellectually and imaginatively, in such a way and to such a degree that he or she is able to provide a presentation of it of just the kind that one of its own adherents would give. It is that even in so doing, the mode of presentation will inescapably be framed within and directed by the beliefs and purposes of one's own point of view. And in recording the history of a conflict...how one recounts that history will depend upon what from one's own point of view one takes to have been the outcome." See also *WJ?* p. 175-176.

constitutes the evidentiary signal--agreement from rival perspectives--which forms the basis of MacIntyre's defense.<sup>22</sup> And so that defense stands in need of reinforcement.<sup>23</sup>

What is required to diminish with actual evidence the threat to the Catholic position on natural law posed by the existence of intractable disagreement about it, is the intellectual testimony of a 2nd-first-language speaker of the Thomistic Aristotelian tradition who was an actual adherent of some relatively stable rival tradition. Dissenting mature members of rival traditions are plentiful; the necessary reinforcement needed for MacIntyre's defense to succeed must take the form of a ladder, of scaffolding, enabling the difficult ascent to maturity into the Thomistic Aristotelianism tradition. But on Aquinas's account, this is the sort of tradition where only communal immersion can

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<sup>22</sup> Here I seem to be interpreting craft-tradition rationality in agreement with Francis Michael Walsh, who in his Sept 2021 *Irish Theological Quarterly* piece, "The Delicate Art of Putting Intractable Moral Disputes Into Question" writes: "A tradition is coherent with itself when it can explain historically what had to be denied and why it had to be denied if its contrary on which the tradition is currently based was to be affirmed. In addition, a tradition is also coherent when it can explain how its social practices are productive of the goods that attain the ends it professes to pursue. Supplying for the missing history or sociology allows a questioner to probe to see how coherently a succeeding tradition of rationality has been in achieving the changes it was introduced to make. In this way, clues to new lines of attack and counterattack may appear, lines that all sides can pursue in the form of further probing questions—*provided all sides are willing to judge the adequacy of the answers of the other side by the incommensurable standards of the other side, and not by their own equally incommensurable standards.* (Emphasis mine.) (<https://doi.org/10.1177/00211400211040752> accessed 11/7/2021.

<sup>23</sup> MacIntyre, IDNL, p. 341: "It is a revealed truth...that the truths of natural law can be known prior to and independently of any revealed truth, including this revealed truth." But if truths of the natural law are interdependent with, nested in, an account of practical rationality, and that account interdependent with a tradition which is not yet known to be rationally preferable to some rival with whom no such truths of natural law cohere, then it would seem an agnostic stance toward those natural law claims would be in order. The mere possibility of overcoming incommensurability is not enough: to make truths about the natural law or any truths actually known, they must rest inside an actually defensible overall account, not a possibly defensible overall account. But in light of what neuroscience and the study of its implications show us about our desperate attachment to our own predictive model, unless we can secure agreement from—actually persuade—a 2nd-1st-language speaker of our own tradition from some rival, about the rational superiority of our own tradition over theirs, we are left with merely possible and not actual persuasion and thus merely possible and not actual overcoming of incommensurability. Failing to thus persuade, there is no reason to suppose our desperately preferred model is to be rationally preferred, and their contradictory perspective not.

support acquisition of 2nd-first language competency in it, acquired in the same way that a first language is acquired: immediately through participation in social life, and not through necessarily unsuccessful attempts at translation. And thus for such rich traditions to come in evidence-producing contact with any other, mature members of each must be grown up first in their home tradition, and then again in the enacted narrative of a socially embodied second tradition,<sup>24</sup> a transition very nearly inaccessible to those who have grown up outside any such tradition, individuals who MacIntyre describes as “citizens of nowhere”<sup>25</sup>. Already the challenge of accelerating this process of such tradition-on-tradition interactions--one which MacIntyre admits may take centuries of attempts<sup>26</sup>--is plain. Describing an accelerator/detector complex sufficient to meet this challenge is the burden of the third and final part of this essay. But first, the task of persuading--first attracting, and then converting--members of some traditions toward others must be sketched; this is the domain of rhetoric, to which we now turn.

## ***II.***

MacIntyre’s craft-tradition defense of Thomistic Aristotelianism and its implied defense of the rational accessibility of the first principles of natural law provides substantial resources to guide what can here be only a sketch of necessary rhetorical resources. First, his account of dialectic throughout his published works provides some detail regarding the tasks to be accomplished in conversion, whether to maturity within the tradition, or conversion from without: the ratcheting up of first principles from lived

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<sup>24</sup> See *WJ?* p. 374-75.

<sup>25</sup> *WJ?* p. 388, 395.

<sup>26</sup> *IDNL* p. 325.

experience, nesting these over time in the context of a broad, consistent and fecund set of principles, the drawing out from these of solution sets to identified problems, defending these problem-resource-solution sets individually and as a whole from the best arguments generated internally and from rival perspectives, while predicting and explaining the emergence and persistence of limits to such success in the best of rival traditions. Each of these areas will have constructions tailored to the backgrounds of those learning the ropes, whether to less mature inhabitants of the home tradition, or to mature members of rivals.

Only in the advancement of the tradition by its mature members does the argumentation proceed from the fullest set of resources. Less mature members of this home tradition do not grasp all the members of that set, and thus must be persuaded by less. Members of rival traditions, even after they achieve second-first-language competency in the tradition in question and thus in some way grasp all of these resources, fail to accept at least some of them; this is precisely why conversion is in question. Thus whether inside a tradition or across its boundary with another, the deployment of rhetoric--tailored specifically to the background of the persuaded--is inescapable. This craft of rhetoric, always appealing to fewer resources than those available to its mature members, may include persuasive tactics that appeal to premises which are accepted by the listener, but which the mature speaker believes to be false. Such persuasion may take place openly--"granting, for the sake of argument", as we sometime say--or without such acknowledgement; if the latter, then perhaps fairly presumed (where the gap in

resources is well known and some oversimplification is expected), or if not fairly, then either before such acknowledgement is due (perhaps a kind of short-lived paternalism), or after (a kind of deception, or hiding.)

The craft-tradition which began from Socratic questioning has since its inception a rich and varied connection with rhetoric in all these forms. In the *Republic*, Plato portrays Socrates as discussing the topic of education, described as a kind of turning of the soul to see and desire rightly, in the context of his persuading Plato's own brothers to care for virtue and learning for their own sake, gently persuading them while hiding from them the fact that they are not healthy.<sup>27</sup> In the *Symposium*, Alcibiades is portrayed as seeing through Socrates' incessant paternal care for his soul at every opportunity, recognizing the purpose of the rhetorical craft Socrates tirelessly but perhaps in the end unfruitfully employs.<sup>28</sup> In the *Phaedo*, Socrates is shown to be a master craftsman of rhetoric, gently moving through one strategy in argument to another in a way that both appeals to and transforms the minds and hearts of his dearest friends on the most important of topics and on the most momentous of occasions.<sup>29</sup> Readers of MacIntyre's 2016 essay on desire, practical reasoning and narrative, written amidst his eighth decade of life, may recognize in its steady guidance of plain, unphilosophical persons through the complex of issues characterizing modern social life, this same gentleness of

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<sup>27</sup> See Plato's *Republic*: "gently persuading him while hiding from him the fact that he's not healthy" (476e). A defense of this interpretation can be found in my "Dialectic and a Realist Informed-Desire Account of the Good", paper #6 in my *Toward Computational Natural Law* (<https://tinyurl.com/TowardComputationalNaturalLaw>), in preparation, accessed 11/10/2021.

<sup>28</sup> See Plato's *Symposium*, especially Alcibiades' speech culminating at 221e ff.

<sup>29</sup> See Plato's *Phaedo*, especially Socrates's behavior as characterized in *Phaedo's* interlude at 89a

spirit.<sup>30</sup> Among the rhetorically effective strategies employed there is the deep attention to the insights and limitations of adherents to rival perspectives along this path, exhibiting rigor, sympathy and impartiality that both inspires confidence in the analysis and quietly builds trust in the reader.<sup>31</sup> This barest mention of the role of rhetoric in the craft-tradition from its inception to its best current expression can only promise and remind us of its deep rootedness in craft-tradition approach to education. It suggests grounds for hope that rhetorical tool sets have been and can be developed for attracting and persuading mature members of rival traditions through dialogue. Calling to mind these resources can instill hope that the evidentiary signal to which MacIntyre's defense of Aquinas's teaching on the natural law has led us to recognize can in fact be produced.

Yet it remains true that the kind of dialogue necessary for the production of that signal rarely occurs, requiring considerable time and quite specific sets of circumstances. We turn next to an overall framework for deploying such rhetoric toward the accelerated production and resolution of such discussions, so that such dialogue may require less time and more commonly available circumstances.

### III.

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<sup>30</sup> *ECM*, p. ix - xii and throughout, esp. 166 - 167.

<sup>31</sup> *ECM*, especially MacIntyre's treatment of Williams, the core of which extends from p. 150 through 165, and beyond.

## Toward a Computational Natural Law Defense of the Sanctity of Human Life

<https://tinyurl.com/dCEC21-Loughran> | <https://tinyurl.com/dCEC21-Loughran-comments>

Detectors are designed to transform a signal produced by some behavior of interest into a form that can be recognized and used by human beings. Output from detectors that are produced by the behavior of interest are known as signal; other output produced by other factors are known as noise. Signal must be produced at a sufficient rate and signal/noise ratio to justify the cost of detector construction. In many cases, signal must be enhanced. In the age of COVID, more of us than ever are aware of how PCR works: a thermally-cycled and catalyzed chain reaction replicating a small bit of DNA until sufficient quantities are produced to be detectable at an affordable price. Sometimes it is the signal itself, causally downstream from the behavior in question, that is enhanced (such as in a photomultiplier tube); other times it is the behavior itself that is magnified (such as increasing the luminosity and thus the number of collisions--the behavior of interest--in a particle accelerator. Often, both strategies are deployed together, where both the behavior in question and its downstream effects are both accelerated. (At the LHC, increasing luminosity in the accelerator increases the investigated behavior; detectors such as CMS and Atlas further amplify the downstream signal.) It is expensive to hold a collaboration together to pursue some common line of investigation, and so increasing the rate of signal production is often part of the practical requirements for successful inquiry.<sup>32</sup> How then might we accelerate the interaction of traditions and amplify the signal produced thereby?

To overcome the challenge of relativism concerning the natural law, the behavior of interest is the agreement, for different sets of reasons, of a pair of members from rival traditions, each a first-language (thus, adequately informed) speaker of both their own and that rival tradition

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<sup>32</sup> As MacIntyre notices in response to Clairmont: "...it can also on occasion be important not to move too slowly, something that often signals to the other party a lack of commitment to the conversation." *IDNL*, p. 224.

concerning the dispute in question. In such cases, agreement indicates that conclusions are warranted by the actual information in question, and not merely by one particular way that information can be stably processed, when another way is available that takes account of the same information and yet reaches a different conclusion. Agreement after dialogue by persons from rival backgrounds is preliminary empirical evidence that one's assent is due not to an accident of one's formation, but to something more.<sup>33</sup> Such agreement is the behavior in question; to increase the rate of its detected presence (if actual) or absence (if not), we can either arrange more of the conditions that produce it, amplify certain of its causal implications, or both.

The conditions MacIntyre has identified in his body of work for the production of such agreement amount to the following. Begin with the idea of persons formed to maturity in their own and one another's traditions (insofar as the latter, as 2<sup>nd</sup> first-language speakers.) Now imagine such persons 1) are brought into contact in pairs; 2) interact sufficiently to recognize this second-first-language competency in one another; 3) explore the two traditions together—call this **interacting over the traditions**—until a mutually satisfying account of each

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<sup>33</sup> On MacIntyre's account, with Augustine and Aquinas, that "something more" is correspondence with external reality: *WJ?* P. 356; *3RV* p. 100 (from Augustine) and 125 (from Aquinas); *IDNL* p. 21. But even an account of inquiry that recognized movement away from less fecund accounts but not toward any particular one, such as Thomas Kuhn's, would suffice to raise the question as to whether some emerging tradition might have the resources to show participants of another on their own terms that theirs is being left behind, shaken out of contention on the path toward useful and stable world views, by thus-far-evident lack of resourcefulness. Hence, the "something more" that is indicated by the possibility of one tradition overcoming another is left unspecified in the argument here. In a forthcoming paper ("Expressivist Creation"), I consider the question of whether such agreement from multiple perspectives need be explained by information produced externally to our consciousness—by some external world to which our judgments, when true, correspond; or instead perhaps just to the nature of rational perspective exchange itself, where the formation of one mind can correspond to the formation of another mind—to that same mind, and others like it, who are adequately informed, such that no further formation would change their choices of one perspective over its rival—rather than corresponding to some extramental reality.

member's home tradition can be articulated, and 4) are confronted with the opportunity to recognize and adopt one of the two traditions as having the better case. The nature of this interaction over traditions could and should be explored further. At the least, it would entail, first, agreement from each, about what mature members of the home tradition would recognize as the main lines of assent and practice within each tradition; call this **resource identification**. Second, some account would be offered (again, from each home tradition's perspective) of the problems/challenges overcome with the resources amassed within each; and some account of remaining/emerging problems<sup>34</sup> and challenges facing each; call this **problem identification**. Third, some account of what it looks like now that certain of these challenges have been overcome, and what it might look like when remaining challenges are tackled successfully; call this **solution identification**. These components might be naturally assembled as problem-resource-solution (PRS) triplets<sup>35</sup>, where the dependency of solutions to identified problems on sets of resources are tracked. Two individuals formed to maturity in each of a pair of traditions, though in different order and thus perhaps with different allegiances, might be said to have interacted *thoroughly* over the pair if they produced together two sets of such triplets that each mature member found full as a description of each tradition, where full is less than exhaustive but sufficient in the views of both to fairly characterize the tradition in its promise and appeal; this would of course be a living process, open to recursive revision, but could achieve stability in a manageable period of time, perhaps a full-time year or a small

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<sup>34</sup> *WJ?* p. 167: each such tradition "has within itself ... a more or less well-defined problematic..."

<sup>35</sup> See *IDNL* p. 34.

handful of years.<sup>36</sup> The plausibility of that time estimate improves, if the interacting mature members are both leaders of small teams of graduate students or post-docs, the team of them comprising a laboratory unit. The emerging product then appears to be a book-length project spinning off a series of dissertation- or similarly-sized research investigations into detailed exploration of certain areas of controversy or promise, all under the broader research program of bringing two traditions into evidentiary interaction.

Once both members (or teams of members) agree on characterizing the traditions in question at this challenge-resource-solution level, both are now in substantial contact with the information stored in one another's traditions, and can be said to have completed the first of two parts of interaction over these traditions. (This is of course a revisable state of agreement-thus-far.) For now that they agree about what the traditions have to offer, they can begin to discuss whether and why they accept, or decline to accept, what each tradition has to offer. Again the discussion runs until each member has stated what he/she accepts and why, and clarifying questions are exchanged until neither member supposes that further exchange is likely to affect the outcome (again, a revisable assessment).

During this process any tradition-paired rhetorical strategies which either tradition has developed would be deployed, each considered in full transparency, known to each member in their fullness. A tradition may have resources designed to "persuade...while hiding from (the

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<sup>36</sup> That these interactions will be substantial is thus granted, but that they may require the decades or centuries of interaction that MacIntyre grants to Clairmont in *IDNL* (p. 325) is not and should not be granted, in the practical order.

one persuaded) the fact that they're not healthy", as Socrates is depicted by Plato in persuading Plato's brothers<sup>37</sup>, or Diodotus is depicted by Thucydides as employing to persuade his fellow Athenian citizens not to sack Mytilene.<sup>38</sup> Such hiding presumably has no place in the interaction at this fully-informed level, though it may elsewhere in the process of information exchange.<sup>39</sup> Still, in full disclosure all persuasive resources would be among the problem-resource-solution sets mutually considered. Among such materials are those which MacIntyre describes as explanations to members of rival traditions of how certain problems recognized by that rival tradition were destined to arise and remain unsolved. But there are likely to be others, including techniques for pointing out insensitivity to some kinds of considerations, such as might be produced by the kind of upbringing advocated by a given tradition,<sup>40</sup> and those aligning with the therapies Galef recommends in *Scout Mindset*.

When such interactions have run their course and both mature members return to their home traditions still committed to the home traditions from which they emerged, such a return constitutes preliminary evidence that any remaining disputes between their traditions are thus far substantially intractable<sup>41</sup>. It remains possible that either or both traditions will later develop

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<sup>37</sup> *Republic* V, 476e

<sup>38</sup> Thucydides, *History of the Peloponnesian Wars*, Book III, Ch. 11: "Plain good advice has thus come to be no less suspected than bad; and the advocate of the most monstrous measures is not more obliged to use deceit to gain the people, than the best counselor is to lie in order to be believed."

<https://www.gutenberg.org/files/7142/7142-h/7142-h.htm#link2HCH0011> accessed 8/6/2022.

<sup>39</sup> So, for example, the applicability of a model to data may be introduced prior to the introduction of limitations on that model. Simply ordering information for maximal impact is not thereby hiding that order; it becomes hiding after presentation of subsequent information or the impact of order of presentation is somehow due, but intentionally not provided.

<sup>40</sup> A litany of examples is provided throughout Ibrahim X. Kendi's *How To Be An Antiracist*, One World, 2019.

<sup>41</sup> MacIntyre recognizes and insists on this possibility: "Nothing that I have said goes any way to show that a situation could not arise in which it proved possible to discover no rational way to settle the disagreements between two rival moral and epistemological traditions, so that positive grounds for a relativistic thesis would

in ways that changes the minds of one or both of this pair of the mature members who have interacted over this pair of traditions; in that sense, these interactions are ongoing, and any signal they produce is tentative, a thus-far-considered result. Yet there is a completeness of the interaction, a full-information interaction between the traditions-as-thus-far-constituted, a resting of the case. Call any such results, a **thus-far-intractable outcome**.

Beyond this first condition of such return to home traditions substantially unchanged, two additional identifiable and evidentiarily-significant results are possible. A second possible outcome of such interaction over traditions is of course a change of allegiance in one member, and confirmation in the other, of one of the two interacting agents. In such a case, the resources amassed by one tradition come to seem more attractive to both reviewing agents, from their differently-informed full perspectives, and thus perhaps for different reasons. Call any such results, a **thus-far-settled outcome**. There are, of course, two versions of such outcomes in any interaction over a given pair of traditions: one in which one tradition seems victorious; and another, its rival.

A third outcome looks like the first, but is not identical with it from an evidentiary perspective. In such cases, the mature members will find themselves unable to complete the tasks involved: either to characterize the commitments, challenges and solutions of either or both traditions in the pair, or to settle on a preferred outcome if they succeed in characterizing them. This is different from a thus-far-intractable outcome: in intractable disputes, the informed members

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emerge. But this I have no interest in denying. For my position entails that there are no successful *a priori* arguments which will guarantee in advance that such a situation could not occur.” MacIntyre, AV2nd, p. 277.

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agree that they have assessed the traditions fully, and agree that they disagree about which of the traditions has a strong enough case to attract allegiance. But in this third type of case, what the interacting agents agree upon is that their progress is insufficient, and perhaps will remain insufficient to generate such disagreement. They do not disagree about which tradition is to be preferred over the other, yet; they instead agree that they can find no such grounds, so far, for such disagreement, either because they cannot agree upon how to characterize one another's traditions, or upon individual verdicts if they can. Perhaps these traditions are two different ways to look at the world, with neither having a stronger case; not rivals, but complements, or perhaps they are just harder to characterize than the members (and perhaps their teams) can so far accomplish. Either way, call this a **thus-far-unresolved outcome**.

This third, unresolved outcome is a source of confusion, and thus, as Socrates would have told us, an occasion for learning. And thus it sets the stage for a fourth kind of outcome, one in which this confusion is seemingly resolved: either to an intractable outcome of the first kind; or to a thus-far-settled agreement of the second kind; or perhaps (moving beyond the confusion of the third) to some advancement recognized (perhaps for different reasons) by the interlocutors as an *emerging synthesis* between the rivals. Call this a **thus-far-transformed outcome**.

There may be other outcomes, such as a perhaps unlikely double conversion of the interacting agents, where each leaves the interaction convinced of the truth of the rival position, perhaps a rare species of the intractable outcome. But these four main categories of outcome suffice as a characterization of the evidentiary signals which might be detected. There are many other ways

that interactions might fail to become evidentiary, remaining in some way incomplete. But these four might become accumulated evidence that a dispute is 1) deeply **intractable**; 2) able to be **settled**; 3) deeply **confusing** (but not yet productively); or 4) productively confusing and thus **transformative**, illuminating and ultimately unifying.

Having examined more closely the evidentiary signal to be produced and some supplementary considerations regarding doing so effectively, let us now work through an algorithm for producing and processing such a signal at a useful and affordable scale.

(1) Begin by **assembling a group of people who mutually identify as mature members of some tradition**. This can be any group of self-identifying like-minded persons.<sup>42</sup> These must approve of one another as mature members, and representatives from this group must subsequently approve of others who come forward as successful students (on which, see more below.)

(2) **Assist this group to articulate a full range of their common commitments as PRS triplets.**

(Foundational principles of the natural law, such as respect for human dignity, might be included here.) In clarifying these commitments, trained assistants would mine the discussion for clarification of common misunderstandings of their analyses of problems, of resource

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<sup>42</sup> Compare with what MacIntyre calls the first stage, historical contingency, in the development of a tradition: *WJ?* beginning at p. 354. As we proceed to assemble and prepare traditions for interaction, we will recapitulate the stages in development of traditions; if we miss any, our interactions will be misshapen, and the results much less clear. That said, if our error is systematic, we may be able to retrofit the results, allowing some room for improvement and iteration. On this possibility, see the history of the Perkin-Elmer misgrinding of the primary mirror for the Hubble Space Telescope, and the development/deployment of the CoSTAR system to refocus it: [https://en.wikipedia.org/wiki/Corrective\\_Optics\\_Space\\_Telescope\\_Axial\\_Replacement](https://en.wikipedia.org/wiki/Corrective_Optics_Space_Telescope_Axial_Replacement), accessed 11/8/21

propositions, of solutions, and of their history<sup>43</sup>, to serve as distractors in upcoming concept inventory preparation, the next step. In an iterative process, these PRS triplets are used to secure agreement with mature members on whether those who can articulate them--to first order, since this step bleeds into the next--also are welcomed as mature members, or at least mature language speakers (depending on whether or not they assent to these commitments) of the founding group. Universal acknowledgement of this level of competency, or welcome as members, begins to establish a basis for interrater reliability of more the formal testing next to be developed.

**(3) Codesign with them a community concept inventory, CCI, with a passing mark for basic first-language competency.** This inventory will be built around both PRS triplets and associated misconceptions of note, including best practices in Concept Inventory development: as to type of exam (typically multiple-choice); number of questions; responses; distractors, etc.<sup>44</sup> Then test, edit, and repeat (TER). This TER process evolves into regular iteration--back-and-forth--among mature membership identification, PRS triplet development, and concept inventories.

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<sup>43</sup> The histories of the development of solutions to problems and the gradual introduction of additional resources for doing so are often left out of concept inventories. But they are not left out of the education of scientists; physicists in particular are remarkable for the mature grasp they show of the history of their discipline, even if it is primarily the rungs on the Nobel Prize ladder.

<sup>44</sup> See, for example, Rebecca S. Lindell, Elizabeth Peak, and Thomas M. Foster, "Are They All Created Equal? A Comparison of Different Concept Inventory Development Methodologies", AIP Conference Proceedings 883, 14-17 (2007) <https://doi.org/10.1063/1.2508680> accessed 11/8/2021. See also Evaluation Instruments: The FCI and Beyond (AMTA: <https://www.modelinginstruction.org/effective/evaluation-instruments/>, accessed 6/1/2023.)

Concept inventories are used regularly in science education. They are developed through a think-aloud protocol—a steady background buzz of trained interlocutors nonjudgementally asking participants why they did or said what they did or said—with mature members of the community.<sup>45</sup> A concept inventory will always involve distractors, which capture important misconceptions. It will have strong face value appeal, meaning that mature members of the community read the questions and acknowledge that those are just the right questions. In physics, the very successful Force Concept Inventory (FCI) functions this way for the community of those who understand Newtonian mechanics. Concept inventories at their best align with additional indicators, such as a set of performance indicators. In Newtonian mechanics, the FCI is aligned with the Mechanics Baseline Test (MBT).<sup>46</sup> Here the intuition is that if anyone really understands the basic qualitative concepts the way that mature practitioners say they must if they can answer the FCI questions, then they ought also (presuming modest mathematical skills) to be able to solve some basic quantified physics problems, which the MBT presents. Reasonable application of a backward-designed curriculum<sup>47</sup> (to peek ahead for a moment) ought to produce gains against both of these indicators. Such measured progress would support triangulation—mutual alignment—of now at least three indicators: documented acceptance by mature members; a mastery score on a concept inventory; and demonstrable success in concept application, starting with application tests. To concept application success at

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<sup>45</sup> Klymkowsky M.W., Garvin-Doxas K. (2020) Concept Inventories: Design, Application, Uses, Limitations, and Next Steps. In: Mintzes J., Walter E. (eds) *Active Learning in College Science*. Springer, Cham. [https://doi.org/10.1007/978-3-030-33600-4\\_48](https://doi.org/10.1007/978-3-030-33600-4_48) accessed 11/8/2021.

<sup>46</sup> See “The Effects of a DLSC Approach on Students Conceptual Understanding in an Undergraduate Introductory Physics Lab” Riaz, Muhammad; Marcinkowski, Thomas J.; Faisal, Ali *EURASIA Journal of Mathematics, Science and Technology Education*, v16 n2 Article em1813 2020

<sup>47</sup> See Wiggins and McTighe, *Understanding By Design*, 2nd, ASCD, 2005.

testing levels might be added—if the tradition is sufficiently mature—peer-reviewed acceptance of papers describing successful concept application to produce results not only new to the student (as tests do) but new to the field. Together, alignment among these measures begins to demonstrate coherence and stability in the assessment system.

At the end of the third stage, there becomes a fact of the matter regarding who is a mature member of which community, what and whether they believe, and thus an identification of first language competency in a given tradition. However it is that the community living out a tradition identifies new members, those prospective members can interact with a representative team of mature members and pass their community concept inventory as well as some subsequent application test. These indicators, taken together, would serve both as a filter for recognizing new mature members, as well as an occasion for fresh tuning of the inventory and application exams for alignment with the judgments of existing mature members about the level of maturity of new candidate members.<sup>48</sup>

(4) Such maturity may come at the end of some course of studies, for which there will be some curriculum. To **backward-design, write and gain mature member approval of a curriculum that suffices to produce persons who can pass the CCI and related exams** is the fourth step in

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<sup>48</sup> This account of achieving maturity in a tradition is necessarily oversimplified here. Certain kinds of tradition require immersion and preliminary acceptance of fundamental tenets in order to gain experience necessary to interpret even its basic practices. See *3RV*, p. 98-99 and *WJ?* P. 396. But the conversion signal needed might be achieved even on the lower extreme of such immersion. Traditions with higher barriers to full maturity can place a higher premium on the judgment of mature members, than on concept inventories or other such tests, which for them will only capture entry level competency (as achievement of a first black belt does in many martial arts traditions. See, for example, <https://www.edge-ma.com/blog/black-belt-what-it-means-and-what-it-doesnt/>, accessed 11/8/2021.

accelerator preparation. Such curriculum preparation involves the identification of key teaching resources, alignment with problem-resource-solution triplets, with the community concept inventory and with any application testing. Emerging through and important to the construction and TER process of such a curriculum will be the identification of person-types. Early use might suggest a simple theory of person types<sup>49</sup>, and then tests can be designed which will identify such types with statistical significance. Eventually intelligent tutoring systems can be added; these are designed with machine learning on large data sets now; further down the road, it is likely that SNP Chips (tests for  $10^2$  to  $10^3$  single nucleotide polymorphisms, or ways in which DNA actually differs among human beings) will be designed to identify and type polygenic behavioral response to curricula; this is an additional enhancement that instructional technology is exploring now.<sup>50</sup>)

So far, all of these steps have set the stage; it is all design and construction of a detector/accelerator complex for the interaction of traditions. This sort of approach to curriculum design would be recognizable to anyone familiar with instructional design.<sup>51</sup> While the idea of a concept inventory suggests an especially intellectual component of the curriculum,

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<sup>49</sup> On types of persons coming to a tradition, whether by types of tradition or types of individuals, see *WJ*? p. 398.

<sup>50</sup> Morris, T. T., Davies, N. M., & Davey Smith, G. (2020). Can education be personalised using pupils' genetic data?. *eLife*, 9, e49962. <https://doi.org/10.7554/eLife.49962>; Shero, J., van Dijk, W., Edwards, A., Schatschneider, C., Solari, E. J., & Hart, S. A. (2021). The practical utility of genetic screening in school settings. *NPJ science of learning*, 6(1), 12. <https://doi.org/10.1038/s41539-021-00090-y> accessed 11/8/2021

<sup>51</sup> Billions of dollars are spent each year encouraging educational companies to innovate in the instructional technology space. I have reviewed over 40 small business innovation research funding proposals in instructional technologies for one federal agency, and another 10 for an industry association. The techniques envisioned here begin to align with best practice in instructional design, and there are no doubt additional best practices which experts in the space could bring to bear.

[https://edtechevidence.org/wp-content/uploads/2021/07/FINAL-K12-EdTech-Funding-Analysis\\_v.1.pdf](https://edtechevidence.org/wp-content/uploads/2021/07/FINAL-K12-EdTech-Funding-Analysis_v.1.pdf) accessed 11/8/2021

there is nothing inherently academic about it. Maturity in a zen motorcycle club might be forged through a series of rides in a master-apprenticeship program<sup>52</sup>; such training too can be backward designed, interrater reliability established, and aligned with a concept inventory.

Whatever the preparation process, once the design is completed for a testable prototype, the next stage is to run it to produce mature members of a single tradition.

(5) **Produce graduates of the curriculum:** Invite students, run the program, and send graduates from that curriculum for inspection by mature members, and for testing with the community concept inventory. As always, test, edit, repeat. Once a prototype is completed and test run, and subsequently TER alignments are made, a statistically significant sizing is targeted and an initial run is executed. Now the accelerator is beginning to produce mature members for some single tradition at scale. Let us pause and note that participation in this process thus far might well be a very good set of developments for a community: an initial set of mature members are identified, its commitments articulated, a measurable sign of maturity developed, a curriculum designed to bring members to maturity, and the process strengthened to grow members at scale.<sup>53</sup> Traditions neither begin nor advance without people coming together around some common commitments; this fifth step spins that cycle up.

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<sup>52</sup> <https://www.hotcars.com/24-things-to-know-before-joining-a-motorcycle-club/> accessed 11/8/2021

<sup>53</sup> There is plenty of advice around for starting or growing communities; see, for example, <https://communelifeblog.wordpress.com/2017/07/03/so-you-want-to-start-a-community/> and <https://medium.com/the-happy-startup-school/seven-seeds-for-growing-a-community-that-people-love-bcb9c3636bda>, accessed 11/8/2021

(6) Once sufficient scale has been achieved for a single tradition, for the full first five steps of the process, **repeat (in series, in parallel or both) for multiple accounts of maturity, multiple communities**. What is being grown, now, is a incept network of rich communities, with some degree of a shared craft of development to scale.

(7) Once multiple communities are prepared, the next step is to move toward interaction. In this seventh step, the collaboration will **pilot a full interaction between a selected pair of communities**. The two-valved heart of this preparation involves, **a)** the exchange of mature members, sent as representatives by the consensus of the mature members gathered in step one, for training toward 2<sup>nd</sup>-1<sup>st</sup>-language competency. A full curriculum and measurement regime is already prepared and validated for this purpose; it may take some time (perhaps 1 - 3 years) to run.

Once these members are certified in one another's traditions as well as in their own, **b)** they begin interacting over their traditions, aiming to produce one of the evidentiary outcomes sketched above (disagreements are intractable, settled, unresolved, or transformed.) In a pilot exchange the number of pairs will be small (perhaps 1 – 3.) The exchange would gain efficiency beyond the pilot, but it should not be forgotten that these members-in-training have already passed through a full training and certification process in their home tradition. This 7<sup>th</sup> step might be accomplished in cohorts of pairings even between a first pair of traditions, so as to magnify the evidentiary signal even in the pilot interaction<sup>54</sup>. As they are produced and agreed

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<sup>54</sup> Eventually this magnification will reach toward statistical significance, but the numbers required for such significance will be a function of the number of variables that emerge through interaction in early pairings.

upon as the evidence produced thus far through their encounters, the contending communities and their convening institution might co-publish the results, and each determine how next to advance their interaction in light of the results thus far. This pilot in turn becomes the best standard thus far for exchange of information between rich communities.

(8) After a pilot interaction is fully played out (or perhaps in a 2<sup>nd</sup> start staggered behind the pilot), work begins to repeat with a larger schedule of pairings—any comer with every comer is the theoretical ideal, but some initial representation of that ideal will have to be selected and run. Eventually, as person-types emerge from the data generated by the first 7 steps of training and testing, a statistically significant number of person-types from each tradition must engage each person type from each rival tradition, in a longer second part of this eighth stage. Subsequent planning based on formative assessment will respond to what is learned through both parts of this eighth stage.

This process through the eighth or ninth stage might involve training hundreds of persons or even a few thousand, but not tens of thousands; results can be achieved over years or even a decade, but will not require a century.<sup>55</sup> What is emerging well within a generation is a productive network of rich communities. Each is rich, insofar as it processes the experience of many people of like mind, refining their shared world views over time in light of new input, in something like a parallel processing environment. Moreover, the emerging network

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Eventually prediction of the number of such variables might become the product of a theoretical science of community interaction, such as assembly theory—see Sarah Walker, *Life As We Know It*—and my 2022 paper presented to the International Society for MacIntyrean Inquiry (“Four Models of Cultural Exchange...”

<sup>55</sup> Pace Clairmont and MacIntyre, IDNL p. 325.

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systematically fosters interaction between and among communities, not primarily producing heat from misunderstanding and disagreement, but rather light from deep exchange through precise formulation of each community's commitments, mutual understanding through curricula validated both against best practices and by the mature members in each community, with 2<sup>nd</sup>-1<sup>st</sup>-language competency measurably achieved through validated tools like a community concept inventory (and perhaps, as the FCI relates to the MBT, others triangulated with it.) This is a network of rich communities assembled and tuned to accelerate the exchange of real understanding, rather than mischaracterization; it is a high-information-flow network of rich communities (hereafter, HIFNRC). It is tuned to listen, observe, and record the outcomes of highly-informed interactions between and among communities. This network is thus an accelerator and detector system.

What will have emerged when such a system is fully running, I want now to end by suggesting, is a new master science.

At first pass, this complex set of interactions would seem to produce data, but there is only a hint of quantifiability, and no discussion at all of the relationship between this social information-processing inquiry and the full range of academic disciplines, including the sciences. So a case needs to be made that this inquiry is a science at all, and if so, what possibly could be meant by the attribution of "master science" to the inquiry.

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The running of a HIFNRC is first of all scientific activity: there is a fact to the matter about who is in a community, what their commitments are, and whether those commitments endure after a full information exchange, and for how many persons of how many discernable person-types. There would be an observable outcome that may constitute affirmation of one tradition's claims to rational superiority—say, some Catholic natural law tradition's claims involving the sanctity of human life—over dissenting accounts. This disagreement between us would not remain intractable, such a tradition might promise for testing, were you formed to maturity in this perspective; this is a stable truth about you, and about others like you (or even perhaps about any human being of any person-type). Moreover, these claims can be made with increasing precision, both as to their substance and to the extent of their appeal to persons who merely fully and measurably understand them. A group of persons gathered around a system for solving quantifiable problems in an accumulating fashion, meets the criteria of a Kuhnian science.<sup>56</sup> If that system also involves retrospective justification of its progress against the historical accounts of its rivals, aiming toward the development of a single stable relationship between the minds of human beings and the world of which they are a part, that system is both a Kuhnian and a MacIntyrean science.<sup>57</sup>

But is such a network a master science...what would it even be to be a *master* science? At the beginning, such a science—call it for now, at most, a network science of human

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<sup>56</sup> Thomas Kuhn, *The Structure of Scientific Revolutions*, 2nd ed., University of Chicago Press, 1970

<sup>57</sup> See MacIntyre, 3RV p. 118-119; IDNL p. 37; EPISTEMOLOGICAL CRISES, DRAMATIC NARRATIVE AND THE PHILOSOPHY OF SCIENCE, *The Monist*, Vol. 60, No. 4, Historicism and Epistemology (OCTOBER, 1977), pp. 453-472

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behavior—would seem far from a master science, since its network of inquiry would be simply too small, and its resulting insights too limited, to be of much relevance, let alone play a leadership or even a governing role, with respect to mature scientific disciplines. But its apparent particurity and powerlessness is illusory. This network is a computational engine, computing stable ways for human beings to interact with the world. And like all computational engines, it's garbage in, garbage out: everything depends on input. This particular computational engine will compute results of interest only insofar as the communities making input to it have much of interest to offer. Part of the development of this engine must be a craft of effectively growing communities, inviting them to come to the table, make the necessary preparations for interaction, receive and make good use of the results.

So now imagine that such a craft of effective invitation advances, and the engine begins to have preliminary successes, promoting and illuminating interactions, and attracting communities at scale and over time. And now consider what nutrients begin to stream into its library of problem/resource/solution triplets.

Three noteworthy sources of enrichment will be among them. 1) As communities are encouraged to exercise the craft of articulation of their world views, their perspectives broaden and deepen. Also, 2) among those contributing are communities which have passed through this whole process already and have grown their resources by learning from other communities with whom they have interacted. And 3), suppose further that the first principles and chief deliverances of various scientific disciplines begin to be deposited in various community triplet

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banks, and that among the mature members of various communities are experts in various disciplines, various crafts, of many sorts. Building on these resources and perhaps others, the worldviews emerging from these enriching communities might begin to look more like complete accounts of the world. But they would not be simply philosophical, theological, historical, physical, etc: not any accounts delivered by any single discipline. They would become, as they mature, essentially multidisciplinary narratives. But unlike any disciplinary or multidisciplinary accounts of the past, they would be measurably attractive, or less so, than any rival accounts thus far to emerge. Thus the deliverances of this inquiry are both measurable under increasing precision (as numbers and types of mature members from various communities contend in quantified and theory-guided experiments ), as well as comprehensive (as these contending world views interact and grow to fullness.) The inquiry begins to look like a science, and perhaps a science in a position to draw on models from multiple disciplines to recognize patterns of interconnection and encourage pursuit of those patterns. The network might begin to exert regulative guidance on the inquiry of its own members, much as the High Energy Physics Advisory Panel does for the various subdisciplines in particle physics<sup>58</sup>, or issue guidance on ethical considerations to guide research as do various advisory committees in other disciplines<sup>59</sup>, or perhaps issue grand interdisciplinary challenges, as does the National Science Foundation.<sup>60</sup>

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<sup>58</sup> <https://science.osti.gov/hep/hepap/Meetings/202111>; accessed 11/8/2021

<sup>59</sup> <https://osp.od.nih.gov/scientific-sharing/secretarys-advisory-committee-on-genetics-health-and-society-archives/> accessed 11/8/2021

<sup>60</sup> [https://www.nsf.gov/news/special\\_reports/big\\_ideas/convergent.jsp](https://www.nsf.gov/news/special_reports/big_ideas/convergent.jsp) accessed 11/8/2021

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The mature members of such a group might in some respects resemble other interdisciplinary groups of experts, living at the intersection of traditions: of their own disciplinary traditions; and of the broader tradition of contemporary science, engineering, medicine, and technology. But such members will carry with them more commitments than these: perhaps from a faith tradition, perhaps political commitments, perhaps cultural or ethnic or national commitments; commitments to integrate these area-specific commitments into a whole embodied account; and commitments to test these whole systems through dialectical engagement involving full information exchange through 2nd-1st-language competent mutuality. The interactions among members with such full sets of commitments would not be small in scale or short in duration; they would succeed only insofar as there is some extended personal commitment to their common project. The questions they address and commitments they make--their stock of problem-resource-solution triplets--are of course tradition-dependent and will thus differ from community to community. They might, for example, include the following:

What are the implications of evolutionary game theory for perception?<sup>61</sup> What does the existence of 150,000 columns of computationally-independent neocortical functionality in the human brain imply for perception, for personal identity, or for the nature of consciousness?<sup>62</sup>

What does the linear relationship between the number of neurons in mammalian brains versus measures of intelligence suggest about the place of human beings in the created order?<sup>63</sup> What are the implications of leading theories of consciousness for a 21st-century Roman Catholic

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Christology?<sup>64</sup> What difference does reading the Old Testament as depicting a one-story universe--after seeing beyond inadequate understandings of apocalyptic language in BCE Palestine--have for rethinking eschatological tensions in the documents of the Second Vatican Council?<sup>65</sup> What do the various lines of inquiry suggesting a computational universe imply for physicalism and the reality of space-time?<sup>66</sup> What are the implications of non-integer models of dimensionality for resolving tensions between quantum theory and relativity, and for the nature of conscious beings whose interactions are governed by such mathematics?<sup>67</sup>

Questions like these are asked regularly in interdisciplinary collaborations at contemporary research universities. But they are not asked together by the same group of people, committed to building a common account of the world in light of their best answers thus far, nor to together living out the implications of those answers in a committed pattern of life.<sup>68</sup>

Nor are contemporary interdisciplinary collaborations typically held together by the long-term commitments that scientific collaborations are. Take, for example, the LIGO collaboration, which detected gravitational waves after 100 years of laying theoretical foundation, decades after indirect evidence; and 25 years after initial NSF investment, involving more than 1000 scientists

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<sup>68</sup> *3RV* p. 80: "In moral theory we are always concerned with the question: what type of enacted narrative would be the embodiment of the actions and transactions of actual social life, of this particular theory? For until we answer this question about a moral theory, we do not know what the theory in fact amounts to; we do not yet understand it adequately."

and 250 research institutions.<sup>69</sup> A mature high-information network of rich communities will amount to conversation among a number of scientific communities (who know themselves to be communities in a much richer sense than contemporary scientists do) committed to just such long-term investigations, building on careers across generations. So their highest-level conversations will both bring into existence and take place within a new master science.

#### IV.

Thus through the formation and interaction of such rich interdisciplinary collaborations can evidentiary signal be produced of precisely the sort required to falsify, or not yet not, claims made by rich traditions to possess the resources necessary to secure rational agreement in the face of genuine incommensurability. As MacIntyre argued in his 10th Gifford Lecture, a renewal of the Thomistic or anyone else's tradition of rational inquiry requires the construction of some such forms of social life, including the new form of university proposed here as an accelerator/detector complex for interacting traditions in a new master science. The resources of traditions, particularly resurrected attention to rhetoric and person-and-tradition-specific persuasion, can be deployed to accelerate and amplify the needed signal. What rational inquiry in the craft-tradition has learned through the development of the disciplines--especially from the brain science of learning, associated instructional strategies and technologies, and the lessons from large-scale international accelerator/detector systems--opens a new horizon with a promising model for construction of an empirical master science. Let's build it.

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<sup>69</sup> LIGO gravitational waves first detection press release and historical overview: <https://www.ligo.caltech.edu/LA/news/ligo20160211> accessed 11/9/2021

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